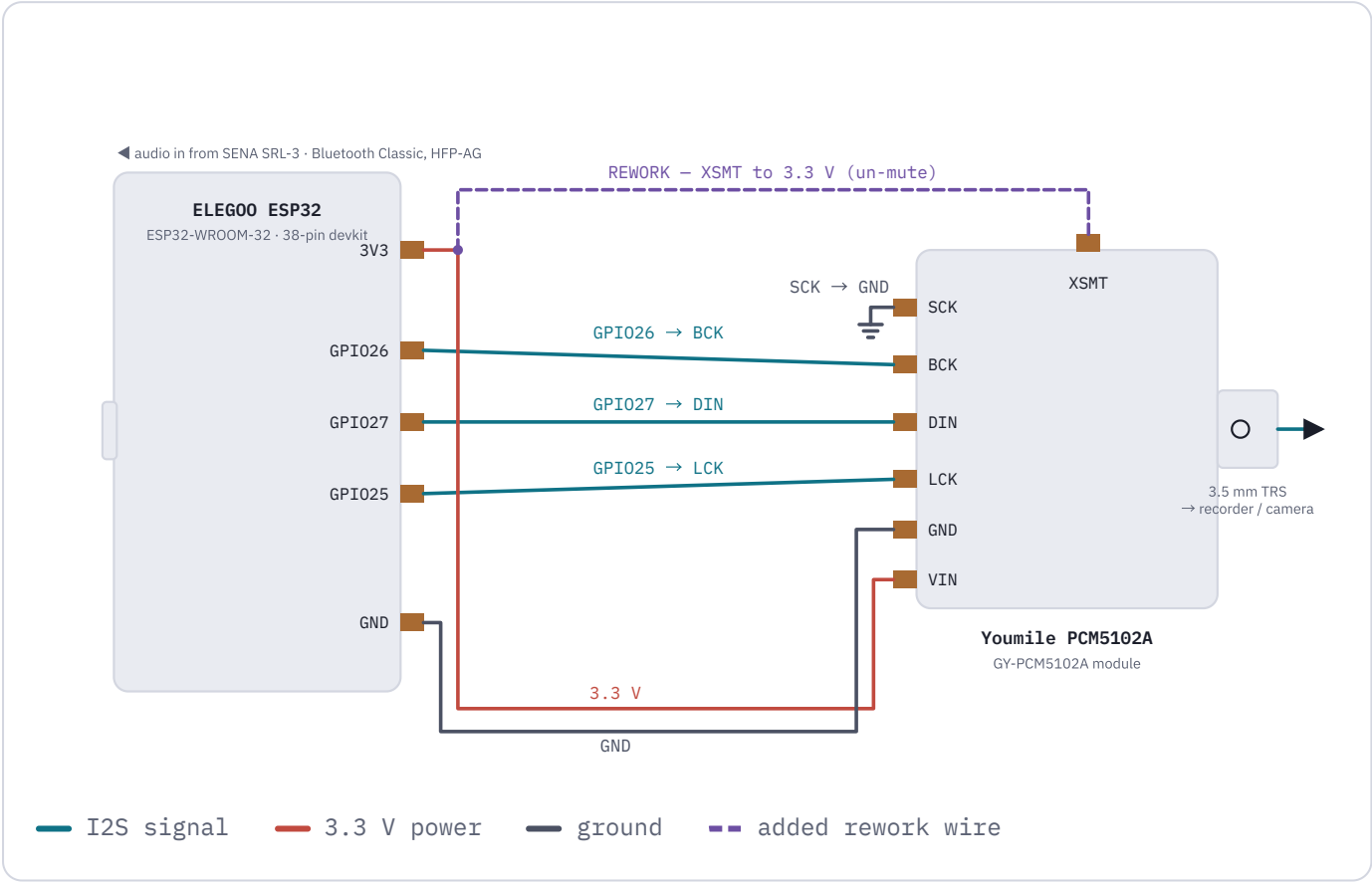
ESP32 → PCM5102A Wiring

Bench wiring for the receive-only intercom bridge: an ELEGOO ESP32 dev board feeding a Youmile PCM5102A DAC, which drives the 3.5 mm analog output. Pin numbers are the ones compiled into `main/app_config.h`.



Three I2S lines carry the received CVSD audio. `SCK` is grounded on the DAC so the PCM5102A runs its own clock (the ESP32 sends no master clock). The dashed wire is the XSMT un-mute fix — needed on this board because its jumper H-rail does not reach the pad.

CONNECTION LIST

ESP32	YOUMILE DAC	SIGNAL	NOTES
GPIO26	BCK	I2S bit clock	256 kHz at 8 kHz / 16-bit
GPIO25	LCK	I2S word select (LRCLK)	—
GPIO27	DIN	I2S data out	ESP32 is the only talker
3V3	VIN	power 3.3 V	or feed VIN from ESP32 5V for more LDO headroom — never both VIN and the 3.3 V pin at once
GND	GND	ground	common reference
3V3	XSMT (top pad)	rework — un-mute	short wire from XSMT to the 3.3 V rail; only because this board's H3L pad has no trace to 3.3 V
—	SCK → GND	tie on the DAC board	selects the internal PLL; no MCLK needed

GPIO32 / 33 / 13 (status LEDs) and GPIO14 (BOND-CLR button) also live in `app_config.h` but are not part of the DAC hookup.

ON THE DAC BOARD — JUMPERS

SOLDER JUMPER	PIN	SET TO	WHY
H1L	FLT	L (GND)	normal latency filter — factory default, leave it
H2L	DEMP	L (GND)	de-emphasis off — factory default, leave it
H3L	XSMT	H (3.3 V)	un-mute — reworked with a wire to 3V3, since the H side of this pad is dead
H4L	FMT	L (GND)	I2S (Philips) format — factory default, leave it
—	SCK	GND	internal PLL from BCK

> **`XSMT` is the one that bites.** Shipped boards bridge all four jumpers to L. That is right for FLT / DEMP / FMT but leaves XSMT low = permanently muted = zero audio. It must read 3.3 V.

POWER

Feed the DAC from the ESP32 `3V3` pin into `VIN` (this is what the repo's wiring table specifies). For more regulator headroom you can instead run ESP32 `5V` into `VIN`. Do not connect `VIN` and the DAC's `3.3V` pin at the same time.

Current draw with an SCO link active is roughly 100–160 mA at 5 V. Power the whole thing from USB or a power bank that tolerates low-current loads.

CHECK BEFORE FIRST POWER-UP

UNPOWERED — CONTINUITY / BUZZER

- [ ] **GPIO26 ↔ BCK** , **GPIO27 ↔ DIN** , **GPIO25 ↔ LCK** — each beeps
- [ ] **ESP32 3V3 ↔ DAC VIN** — beeps
- [ ] **ESP32 GND ↔ DAC GND** — beeps
- [ ] **DAC SCK ↔ DAC GND** — beeps
- [ ] **DAC XSMT ↔ ESP32 3V3** — beeps (the rework wire)
- [ ] **DAC XSMT ↔ DAC GND** — silent (old L bridge fully removed)
- [ ] **ESP32 3V3 ↔ ESP32 GND** — silent (no power short)

POWERED FROM USB — DC VOLTS VS GND

- [ ] **DAC XSMT** ≈ 3.3 V — confirms un-mute
- [ ] **DAC VIN** ≈ 3.3 V
- [ ] **FLT / DEMP / FMT** ≈ 0 V

Then `idf.py build flash monitor`, pair the SRL-3, start a universal intercom, and audio should appear at the 3.5 mm jack when the SCO link comes up.

Pin assignments from `main/app_config.h` (`PIN_I2S_BCLK 26` , `PIN_I2S_WS 25` , `PIN_I2S_DOUT 27`). Behaviour spec: `dummy_intercom_spec.md` rev 0.6. DAC pin order (SCK·BCK·DIN·LCK·GND·VIN) per the GY-PCM5102A module datasheet; verify against the silk on your board before soldering.